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Set Up a Web App Using AWS and VS Code



yakubu abdullahi yusuf

The screenshot shows a VS Code editor with a project named 'NEXTWORK-WEB-PROJECT [SSH: EC2-44-220-231-...]'. The file explorer on the left shows the project structure: 'src/main' (expanded), 'resources', 'webapp', 'WEB-INF', 'index.jsp', and 'pom.xml'. The 'index.jsp' file is selected, and its content is displayed in the editor. The code is as follows:

```
1 <html>
2 <body>
3 <h2>Hello YKB!</h2>
4 <p>This is my Nextwork web application working!</p>
5 </body>
6 </html>
7
```



Introducing Today's Project!

In this project, I will demonstrate how to build a Continuous Integration and Continuous Deployment/Delivery pipeline from scratch. I'm doing this project to learn how to automate the build, testing and deployment of a web application.

Key tools and concepts

In the day 1 of the 7 days challenge project, I used services like ec2 instance and its key pairs, vscode, Apache maven and Amazon coretto 8. The key takeaways include using terminal in vscode, remote SSH, editing application through IDE and terminal

Project reflection

As much as this project is fun, the one thing I didn't expect was the little difference in using MAC and windows in carrying out the project. Using MAC was quite straight forward and easier in terms of the Remote SSH.

This project took me approximately 40 minutes to setup all the services and study the new things I came across. The most challenging part was the Remote SSH, it kept getting disconnected and I had to restart my ec2. It's fun to go through that.



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This project is part one of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project in 24 hours.



Launching an EC2 instance

I started this project by launching an EC2 instance to provide a dedicated and configurable environment for automating the software development and deployment process. The EC2 instance acts as a server where I can install other resources.

I also enabled SSH

SSH is a protocol that provides access to remote servers securely. It allows for the secure connection of our EC2 instances. Once connected, SSH makes sure that the provided private key matches the public on the server.

Key pairs

A key pair is a digital key to a virtual machine. Key pair allows for the secure access of EC2 instances without using passwords. A key pair consists of a public and private key, AWS keeps the public key and the private key is the downloaded key.

As soon as I setup my key pair, AWS automatically downloaded the private key, which is the .pem file. It is very important because it serves as the secure credential for connecting to an EC2 instance via SSH. This file is only provided once.



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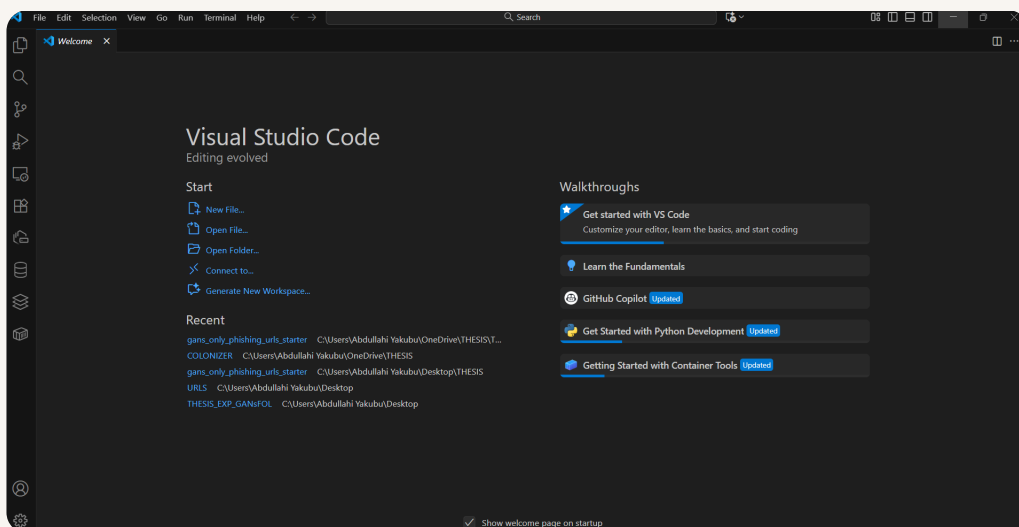
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Set up VS Code

VS Code is a powerful code editor, it is one of the most popular tools for software development. It is a software that allows writing and editing of code. It has an integrated terminal and it also supports many programming languages.

I installed VS Code to create and edit my code, after that, I set up an integrated terminal to get access to my pem file. Using the pem file, I will connect to my EC2 instance remotely, if the private key matches the public stored by AWS.





My first terminal commands

A terminal is also known as a command line, this is where you type texts to send commands to your computer instead of the usual button clicks on the GUI. The first command I ran for this project is `cd "C:\Users\Abdullahi Yakubu\Desktop\LearnNextwork"`.

I also updated my private key's permissions by running the code `chmod 400 keypair.pem`, where `chmod` means change mode, 400 gives just the user the permission to have access to the file and `keypair.pem` is the name for my .pem file.

```
yakubuabduallahiyusuf@DESKTOP-08HCGP3:/mnt/c/Users/Abdullahi Yakubu/Desktop/LearnNextwork/seven$ cd "/mnt/c/Users/Abdullahi Yakubu/Desktop/LearnNextwork/seven/Keypairs"
yakubuabduallahiyusuf@DESKTOP-08HCGP3:/mnt/c/Users/Abdullahi Yakubu/Desktop/LearnNextwork/seven/Keypairs$ ls
keypair.pem
yakubuabduallahiyusuf@DESKTOP-08HCGP3:/mnt/c/Users/Abdullahi Yakubu/Desktop/LearnNextwork/seven/Keypairs$
```

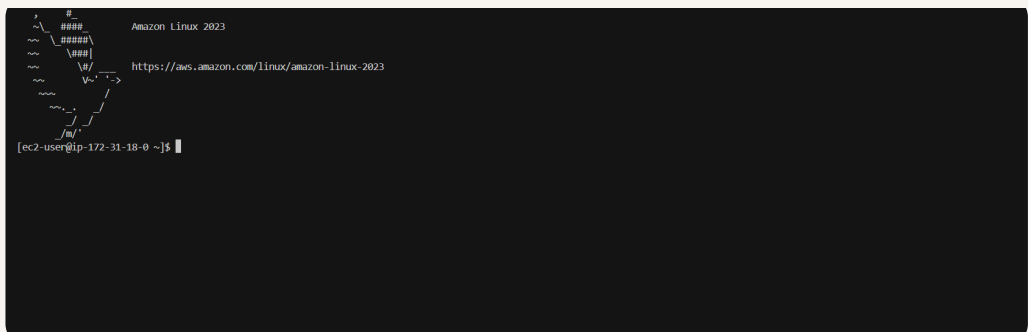


SSH connection to EC2 instance

To connect to my EC2 instance, I ran the command `ssh -i "path to pem file" ec2-user@public IPV4 DNS`

This command required an IPv4 address

A server's IPV4 DNS is the public address of EC2 server that allows devices on the internet to locate and connect to it. I will be using this address to connect my laptop to my EC2 instance for this project.



```
#
##### Amazon Linux 2023
#####
#####
##### https://aws.amazon.com/linux/amazon-linux-2023
#####
#####
#####
#####
#####
[ec2-user@ip-172-31-18-0 ~]$
```



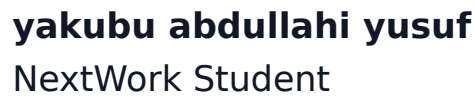
Maven & Java

Apache Maven is a tool for automation and management, which is mostly used for java projects. It has a huge impact in helping developers manage projects in building, dependencies, packaging, linking and also making reports in a standard way.

I need Maven in this project for build automation, compiling, linking, packaging and testing of my applications, which will be fun, stay tuned.

Java is a high level programming language and software platform that is use for building applications ranging from mobile applications up to enterprise level applications. It is used in building applications that can run on almost all devices.

In this project, I will be using java for building my applications along side maven.



In this step I installed Remote - SSH in VS code so I can connect and work on other computers or servers, for example, the EC2 i created.

After creating and adding a New SSH host, the configuration file contains a host, which is like an ALias for my host, the HostName, which is the actual hostname or public DNS of my EC2 instance, the IdentityFile and the User.

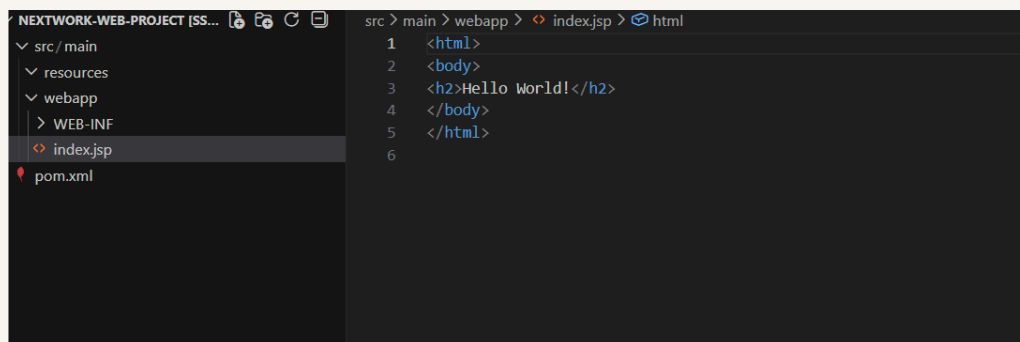
[illegible]



Create the Application

Using VS Code's file explorer, I could see the application's files and folders i downloaded earlier, which are The src, which holds all the files and folders, like the webapp, resources and pom.xml

Two of the project folders created by Maven are src and webapp, which holds all the source code files that define how your web app looks and works and webapp folder contains the web app's files e.g. HTML, CSS, JavaScript, and JSP files.



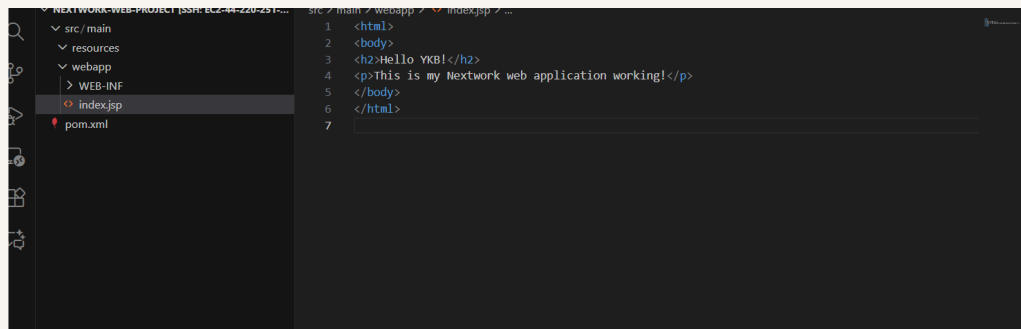
```
src > main > webapp > index.jsp > html
1  <html>
2  <body>
3  <h2>Hello World!</h2>
4  </body>
5  </html>
6
```



Using Remote - SSH

The index.jsp is a similar file to HTML, it is used in Java web apps that allows the generation of dynamic content based on the input of users or data exchange to and from databases.

I edited index.jsp through vscode IDE, which I connected vscode directly to ec2 instance to allow me to edit my application.



```
1 <html>
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